Harder Substitution

Exam code: 9MA0

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Help me understand

Harder substitution

What is integration by substitution?

- Make sure you are familiar with Reverse Chain Rule and Substitution (RCR)
- In more difficult questions the substitution will be given to you

e.g. USE THE SUBSTITUTION $u = x + 3$ TO FIND $\int \left(\frac{x}{x+3}\right)^3 dx$.

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How do I integrate using a given substitution?

- The process is very much the same as before but expect the algebra to be trickier

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

STEP 1: DIFFERENTIATE AND REARRANGE

$$u = x + 3$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$x = u - 3$$

SPOT YOU MAY NEED THIS FOR THE NUMERATOR

STEP 2: REPLACE ALL PARTS OF THE INTEGRAL

$$\int\left(\frac{x}{x+3}\right)^3 dx = \int\left(\frac{u-3}{u}\right)^3 du$$

ALGEBRA MORE AWKWARD!

$$= \int (1 - 3u^{-1})^3 du$$

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USE BINOMIAL EXPANSION

$$= \int \left(1 + 3(-3u^{-1}) + 3(-3u^{-1})^2 + (-3u^{-1})^3\right) du$$

$$= \int (1 - 9u^{-1} + 27u^{-2} - 27u^{-3})\, du$$

STEP 4: INTEGRATE, SUBSTITUTE x BACK IN

$$= u - 9\ln|u| + \frac{27u^{-1}}{-1} - \frac{27u^{-2}}{-2} + c$$

$$= x + 3 - 9\ln|x+3| - \frac{27}{x+3} + \frac{27}{2(x+3)^2} + c$$

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- STEP 1: Differentiate the substitution and rearrange
  - $\frac{du}{dx}$ here can be treated like a fraction (eg × **dx** to get rid of fractions)
- STEP 2: Replace all parts of the integral
  - all **x** terms should be replaced with equivalent **u** terms, including **dx**
  - if a definite integral change the limits from **x** to **u** too
- STEP 3: Integrate and either
  - substitute **x** back in or
  - evaluate the definite integral using the **u** limits
- STEP 4: Find **c**, the constant of integration, if needed

Worked Example

exact value of

$$\int_1^2 \frac{x^2}{\sqrt{1+x^3}}\,dx$$

$$I = \int_1^2 \frac{x^2}{\sqrt{1+x^3}}\,dx$$

$$u^2 = x^3 + 1$$

STEP 1: DIFFERENTIATE AND REARRANGE

$$2u\frac{du}{dx} = 3x^2$$

$$2u\,du = 3x^2\,dx$$

DEAL WITH THIS LATER

$$x^3 = u^2 - 1$$

STEP 2: REPLACE ALL PARTS OF THE INTEGRAL

$x = 1, \quad u^2 = 1^3 + 1 = 2, \quad u = \sqrt{2}$

$x = 2, \quad u^2 = 2^3 + 1 = 9, \quad u = 3$

EXACT VALUE REQUIRED AT END

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"ADJUST" AND "COMPENSATE"

$$I = \frac{1}{3}\int_1^2 \frac{3x^2}{\sqrt{1+x^3}}\,dx$$

$$I = \frac{1}{3}\int_{\sqrt{2}}^3 \frac{2u\,du}{\sqrt{1+(u^2-1)}}$$

$$I = \frac{2}{3}\int_{\sqrt{2}}^3 \frac{u}{\sqrt{u^2}}\,du$$

$$I = \frac{2}{3}\int_{\sqrt{2}}^3 du$$

$$\int_{\sqrt{2}}^3 1\,du$$

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